mathematics

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Some Results on Coefficient Estimate Problems for Four-Leaf-Type Bounded Turning Functions

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Abstract: Let $\mathcal{BT}_{4l}$ denote a subclass of bounded turning functions connected with a four-leaf-type domain. The goal of the study is to probe into the bounds of coefficients $|b_6|, |b_7|, |b_8|$, the bounds of the logarithmic coefficients, and the third-order determinants $|H_{3,1}|, |H_{3,2}|, |H_{3,3}|$ for the functions in this class.

Keywords: Hankel determinant; Schwarz functions; bounded turning functions; four-leaf-type domain

MSC: 30C45; 30C80

1. Introduction And Definitions

Let $\mathcal{H}$ represent a family of mappings of the following kind, denoted by g

$$g(\xi) = \xi + \sum_{n=2}^{\infty} b_n \xi^n \tag{1}$$

in the unit disc $\mathcal{U} = \{\xi \in \mathbb{C} : |z| < 1\}$. We refer to $\mathcal{S}$ as a subfamily of $\mathcal{H}$, which consists of univalent functions in $\mathcal{U}$.

For each function $g \in \mathcal{H}$ and positive integers i and n, the Hankel determinant $H_{i,n}(g)$ is defined by

$$H_{i,n}(g) := \begin{vmatrix} b_n & b_{n+1} & \dots & b_{n+i-1} \\ b_{n+1} & b_{n+2} & \dots & b_{n+i} \\ \vdots & \vdots & \dots & \vdots \\ b_{n+i-1} & b_{n+i} & \dots & b_{n+2i-2} \end{vmatrix} \quad (b_1 = 1).$$

The Hankel determinant $H_{i,n}(g)$ was introduced by Pommerrenke [1,2]. The third Hankel determinants with $n \leq 3$ are as follows:

$$H_{3,1}(g) = b_3 b_5 + 2b_2 b_3 b_4 - b_3^3 - b_4^2 - b_2^2 b_5, \tag{2}$$

$$H_{3,2}(g) = b_4(b_3 b_5 - b_4^2) - b_5(b_2 b_5 - b_3 b_4) + b_6(b_2 b_4 - b_3^2), \tag{3}$$

$$H_{3,3}(g) = b_5(b_4 b_6 - b_5^2) - b_6(b_3 b_6 - b_4 b_5) + b_7(b_3 b_5 - b_4^2). \tag{4}$$

For the first time, the bounds of the third-order Hankel determinant $|H_{3,1}(g)|$ for the families of $\mathcal{S}^*$ of starlike functions, $\mathcal{K}$ of convex functions and $\mathcal{R}$ of functions with bounded turning were investigated by Babalola [3]. Recently, the sharp bounds of the Hankel determinant $H_{3,1}(g)$ of subclasses of analytic functions were obtained by many authors [4–10].

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Citation: Wen, C.; Li, Z.; Guo, D. Some Results on Coefficient Estimate Problems for Four-Leaf-Type Bounded Turning Functions. *Mathematics* **2024**, *12*, 1875. https://doi.org/10.3390/math12121875

Academic Editor: Jay Jahangiri

Received: 14 April 2024
Revised: 7 June 2024
Accepted: 7 June 2024
Published: 16 June 2024

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For $g \in \mathcal{S}$, let $F_g(\xi)$ be defined as the

$$F_g(\xi) = log\frac{g(\xi)}{\xi} = 2\sum_{n=1}^{\infty}\delta_n\xi^n, \tag{5}$$

where $\delta_n = \delta_n(g)$ is referred to as the logarithmic coefficients of g. In the theory of univalent functions, these coefficients play an important role for different estimates. The problem of the best upper bounds for δ_n is still open. In fact, even the proper order of magnitude is still not known. It is known, however, for the starlike functions that the best bound is $|\delta_n| \leq \frac{1}{n}(n \geq 1)$, and that this is not true in general [11].

Differentiating (5), we have

$$\frac{\xi g'(\xi) - g(\xi)}{\xi g(\xi)} = 2\delta_1 + 4\delta_2\xi + 6\delta_3\xi^2 + 8\delta_4\xi^3 + 10\delta_5\xi^4 + \cdots.$$

From (1), we obtain

$$\begin{aligned}\frac{\xi g'(\xi) - g(\xi)}{\xi g(\xi)} &= b_2 + (2b_3 - b_2^2)\xi + (3b_4 - 3b_2b_3 + b_2^3)\xi^2 + (4b_5 - 4b_2b_4 - 2b_3^2 + 4b_2^2b_3 - b_2^4)\xi^3\\ &+ (5b_6 - 5b_2b_5 - 5b_3b_4 + 5b_4b_2^2 + 5b_2b_3^2 - 5b_2^3b_3 + b_2^5)\xi^4 + \cdots.\end{aligned}$$

By comparing the coefficients, we achieve

$$\begin{cases}\delta_1 = \frac{1}{2}b_2,\\ \delta_2 = \frac{1}{2}(b_3 - \frac{1}{2}b_2^2),\\ \delta_3 = \frac{1}{2}(b_4 - b_2b_3 + \frac{1}{3}b_2^3),\\ \delta_4 = \frac{1}{2}(b_5 - b_4b_2 + b_3b_2^2 - \frac{1}{2}b_3^2 - \frac{1}{4}b_2^4),\\ \delta_5 = \frac{1}{2}(b_6 - b_2b_5 - b_3b_4 + b_4b_2^2 + b_2b_3^2 - b_2^3b_3 + \frac{1}{5}b_2^5).\end{cases} \tag{6}$$

In 2022, Sunthrayuth et al. [12] introduced a subclass of bounded turning functions associated with a four-leaf function defined by

$$\mathcal{B}T_{4l} = \left\{g \in \mathcal{S} : g'(\xi) \prec 1 + \frac{5}{6}\xi + \frac{1}{6}\xi^5\right\},$$

and obtained Kruskal inequality, the bounds of the coefficient inequalities and the two-order Hankel determinant of bounded turning class $\mathcal{B}T_{4l}$.

Utilizing the estimates of the coefficients of the Schwartz function, we study the third-order Hankel determinants $|H_{3,1}(g)|$, $|H_{3,2}(g)|$ and $|H_{3,3}(g)|$ for the class $\mathcal{B}T_{4l}$; also, we obtain the bounds for the logarithmic coefficients for $g \in \mathcal{B}T_{4l}$.

Let ω_0 be the family of Schwarz functions. Thus, the functions $w \in \omega_0$ can be expressed as a power series:

$$w(\xi) = \sum_{n=1}^{\infty} c_n\xi^n. \tag{7}$$

Lemma 1 (see [13]). *Suppose a function w is a member of ω_0. Then,*

$$|c_2| \leq 1 - |c_1|^2, \quad |c_3| \leq 1 - |c_1|^2 - \frac{|c_2|^2}{1 + |c_1|}, \quad |c_4| \leq 1 - |c_1|^2 - |c_2|^2,$$

$$|c_5| \leq 1 - |c_1|^2 - |c_2|^2 - \frac{|c_3|^2}{1 + |c_1|}, \quad |c_6| \leq 1 - |c_1|^2 - |c_2|^2 - |c_3|^2,$$

$$|c_7| \leq 1 - |c_1|^2 - |c_2|^2 - |c_3|^2 - \frac{|c_4|^2}{1 + |c_1|}.$$

Lemma 2 (see [14,15]). *Suppose a function w is a member of ω_0. Then,*

$$|c_1c_3 - c_2^2| \leq 1 - |c_1|^2, \quad |c_2c_4 - c_3^2| \leq (1 - c_1^2)^2, \quad |c_3c_5 - c_4^2| \leq 1 - c_1^2.$$

Lemma 3 (see [12]). *If $g \in \mathcal{B}T_{4l}$, then*

$$|b_2| \leq \frac{5}{12}, \quad |b_3| \leq \frac{5}{18}, \quad |b_4| \leq \frac{5}{24}, \quad |b_5| \leq \frac{1}{6}.$$

Lemma 4 (see [11]). *If $w \in \omega_0$, then $|c_n| \leq 1 \quad (n \geq 1)$.*

Lemma 5 (see [16]). *Let $w \in \omega_0$; then, for $\gamma \in \mathbb{C}$, we have*

$$\left|c_2 + \gamma c_1^2\right| \leq max\{1, |\gamma|\}$$

Lemma 6 (see [17]). *If $w \in \omega_0$. Then, we obtain*

$$|c_3 + \gamma c_1c_2 + \varsigma c_1^3| \leq 1,$$

where $(\gamma, \varsigma) \in D_1 \cup D_2$, with

$$D_1 = \{(\gamma, \varsigma) \in \mathbb{R}^2 : |\gamma| \leq \frac{1}{2}, -1 \leq \varsigma \leq 1\},$$

$$D_2 = \{(\gamma, \varsigma) \in \mathbb{R}^2 : \frac{1}{2} \leq |\gamma| \leq 2, \frac{4}{27}(|\gamma| + 1)^3 - (|\gamma| + 1) \leq \varsigma \leq 1\}.$$

Lemma 7 (see [18]). *Let $w \in \omega_0$ and $\gamma \in \mathbb{C}$; we obtain*

$$\left|c_4 + 2\gamma c_1c_3 + \gamma c_2^2 + 3\gamma^2 c_1^2c_2 + \gamma^3 c_1^4\right| \leq max\{1, |\gamma|^3\}.$$

Lemma 8 (see [19]). *If $w \in \omega_0$, then for all $\gamma \in \mathbb{C}$, we have*

$$|c_5 + 2\gamma c_1c_4 + 2\gamma c_2c_3 + 3\gamma^2 c_1c_2^2 + 3\gamma^2 c_1^2c_3 + 4\gamma^3 c_1^3c_2 + \gamma^4 c_1^5| \leq 1.$$

2. The Bounds of the Third Hankel Determinant for $g \in \mathcal{B}T_{4l}$

Theorem 1. *If $g \in \mathcal{B}T_{4l}$, then*

$$|b_6| \leq \frac{5}{36}, \quad |b_7| \leq \frac{5}{42}, \quad |b_8| \leq \frac{5}{48}.$$

The bounds are sharp.

Proof. For a function $g \in \mathcal{B}T_{4l}$, there exists a Schwarz function $w(\xi)$, such that

$$g'(\xi) = 1 + \frac{5}{6}w(\xi) + \frac{1}{6}w^5(\xi),$$

Comparing the coefficients, we yield

$$b_2 = \frac{5}{12}c_1, \quad b_3 = \frac{5}{18}c_2, \quad b_4 = \frac{5}{24}c_3, \quad b_5 = \frac{1}{6}c_4, \tag{8}$$

$$b_6 = \frac{5c_5 + c_1^5}{36} \tag{9}$$

$$b_7 = \frac{5c_6 + 5c_1^4c_2}{42}, \tag{10}$$

$$b_8 = \frac{5c_7 + 5c_1^4c_3 + 10c_1^3c_2^2}{48}. \tag{11}$$

From (9) and Lemma 1, we have

$$\begin{aligned} |b_6| &\leq \frac{1}{36}\left[5|c_5| + |c_1|^5\right] \\ &\leq \frac{1}{36}\left[5(1 - |c_1|^2 - |c_2|^2 - \frac{|c_3|^2}{1+|c_1|}) + |c_1|^5\right] \\ &= \frac{1}{36}\left(5 - 5|c_1|^2 + |c_1|^5 - 5|c_2|^2 - \frac{5|c_3|^2}{1+|c_1|}\right) \\ &\leq \frac{1}{36}\left(5 - 5|c_1|^2 + |c_1|^5\right) \\ &\leq \frac{5}{36}. \end{aligned}$$

From (10) and Lemma 1, we achieve

$$\begin{aligned} |b_7| &\leq \frac{1}{42}\left[5|c_6| + 5|c_1|^4|c_2|\right] \\ &\leq \frac{1}{42}\left[5(1 - |c_1|^2 - |c_2|^2 - |c_3|^2) + 5|c_1|^4|c_2|\right] \\ &= \frac{1}{42}\left(5 - 5|c_1|^2 + 5|c_1|^4|c_2| - 5|c_2|^2 - 5|c_3|^2\right) \\ &\leq \frac{1}{42}\left[5 - 5|c_1|^2 + 5|c_1|^4(1 - |c_1|^2)\right] \\ &= \frac{1}{42}\left(5 - 5|c_1|^2 + 5|c_1|^4 - 5|c_1|^6\right) \\ &\leq \frac{5}{42}. \end{aligned}$$

From (11) and Lemma 1, we obtain

$$\begin{aligned} |b_8| &\leq \frac{1}{48}\left[5|c_7| + 5|c_1|^4|c_3| + 10|c_1|^3|c_2|^2\right] \\ &\leq \frac{1}{48}\left[5(1 - |c_1|^2 - |c_2|^2 - |c_3|^2 - \frac{|c_4|^2}{1+|c_1|}) + 5|c_1|^4|c_3| + 10|c_1|^3|c_2|^2\right] \\ &= \frac{1}{48}\left[5 - 5|c_1|^2 + (-5 + 10|c_1|^3)|c_2|^2 + 5|c_1|^4|c_3| - 5|c_3|^2 - 5\frac{|c_4|^2}{1+|c_1|}\right] \\ &\leq \frac{1}{48}[5 - 5|c_1|^2 + (-5 + 10|c_1|^3)|c_2|^2 + 5|c_1|^4(1 - |c_1|^2 - \frac{|c_2|^2}{1+|c_1|})] \\ &= \frac{1}{48}[5 - 5|c_1|^2 + 5|c_1|^4 - 5|c_1|^6 + \frac{-5 - 5|c_1| + 10|c_1|^3 + 5|c_1|^4}{1+|c_1|}|c_2|^2]. \end{aligned}$$

Setting $c = |c_1|$ and $d = |c_2|$, we obtain

$$|b_8| \leq \frac{1}{48}\epsilon_1(c, d)$$

where

$$\epsilon_1(c, d) = 5 - 5c^2 + 5c^4 - 5c^6 + \frac{-5 - 5c + 10c^3 + 5c^4}{1+c}d^2.$$

The critical points of ϵ_1 satisfy

$$\begin{cases} \frac{\partial \epsilon_1}{\partial c} = -10c + 20c^3 - 30c^5 + \frac{15c^4 + 40c^3 + 30c^2}{(1+c)^2}d^2, \\ \frac{\partial \epsilon_1}{\partial d} = \frac{-10 - 10c + 20c^3 + 10c^4}{1+c}d = 0. \end{cases}$$

Applying numerical computations, we have

$$\begin{cases} c_0 = 0, \\ d_0 = 0, \end{cases} \quad \begin{cases} c_1 = 0.8668, \\ d_1 = 0.7940, \end{cases} \quad \begin{cases} c_2 = 0.8668, \\ d_2 = -0.7940. \end{cases}$$

Thus, in $(0,1) \times (0, 1-c^2)$, there is no critical points which satisfies $0 \leq d \leq 1-c^2$. For $c = 0$,

$$\epsilon_1(0,d) = 5 - 5d^2 \leq 5.$$

For $d = 0$,

$$\epsilon_1(c,0) = 5 - 5c^2 + 5c^4 - 5c^6 \leq 5.$$

For $d = 1-c^2$,

$$\epsilon_1(c, 1-c^2) = 5c^2 + 10c^3 - 5c^4 - 15c^5 + 5c^7 = \eta_1(c) \leq \eta_1(0.7202) = 2.5799.$$

Thus, we have

$$|b_8| \leq \frac{5}{48}.$$

These bounds can be determined by the following extremal functions:

$$g_1(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t^5 + \frac{1}{6}t^{25})dt = \xi + \frac{5}{36}\xi^6 + \frac{1}{156}\xi^{26} + \cdots,$$

$$g_2(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t^6 + \frac{1}{6}t^{30})dt = \xi + \frac{5}{42}\xi^7 + \frac{1}{186}\xi^{31} + \cdots,$$

$$g_3(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t^7 + \frac{1}{6}t^{35})dt = \xi + \frac{5}{48}\xi^8 + \frac{1}{216}\xi^{36} + \cdots.$$

□

Theorem 2. *If $g \in \mathcal{BT}_{4l}$, then*

$$|H_{2,3}(g)| \quad \leq \quad 0.044516.$$

Proof. Let $g \in \mathcal{BT}_{4l}$.

From (8), we receive

$$|H_{2,3}(g)| = |b_3 b_5 - b_4^2| = \frac{5}{1728}\left|16c_2c_4 - 15c_3^2\right| = \frac{5}{1728}|15(c_2c_4 - c_3^2) + c_2c_4|. \tag{12}$$

Utilizing inequality, Lemma 1 and Lemma 2 in (12), we obtain

$$\begin{aligned} |H_{2,3}(g)| \quad &\leq \frac{5}{1728}[15|c_2c_4 - c_3^2| + |c_2||c_4|] \\ &\leq \frac{5}{1728}[15(1-|c_1|^2)^2 + |c_2|(1-|c_1|^2 - |c_2|^2)] \\ &= \frac{5}{1728}[15 - 30|c_1|^2 + 15|c_1|^4 + (1-|c_1|^2)|c_2| - |c_2|^3)] \\ &\leq \frac{5}{1728}[15 - 30|c_1|^2 + 15|c_1|^4 + (1-|c_1|^2)|c_2| - |c_2|^3]. \end{aligned}$$

Let $|c_1| = c$ and $|c_2| = d$, we obtain

$$|H_{2,3}(g)| \quad \leq \tfrac{5}{1728}\epsilon_2(c,d).$$

Consider

$$\begin{cases} \frac{\partial \epsilon_2}{\partial c} = -60c + 60c^3 - 2cd = 0, \\ \frac{\partial \epsilon_2}{\partial d} = 1 - c^2 - 3d^2 = 0. \end{cases}$$

Applying numerical computations, we obtain

$$\begin{cases} c_1 = 0, \\ d_1 = -0.5774, \end{cases} \quad \begin{cases} c_2 = 0, \\ d_2 = 0.5774, \end{cases} \quad \begin{cases} c_3 = -1, \\ d_3 = 0, \end{cases} \quad \begin{cases} c_4 = 1, \\ d_4 = 0, \end{cases} \quad \begin{cases} c_5 = 0.9998, \\ d_5 = -0.0111, \end{cases} \quad \begin{cases} c_6 = -0.9998, \\ d_6 = -0.0111. \end{cases}$$

Thus, there is no critical point in $(0,1) \times (0, 1-c^2)$.

(1) For $d = 0$,

$$\epsilon_2(c, 0) = 15 - 30c^2 + 15c^4 \leq 15.$$

(2) For $c = 0$,

$$\epsilon_2(0, d) = 15 + d - d^3 \leq \epsilon_2(0, \frac{\sqrt{3}}{3}) = \frac{2\sqrt{3} + 135}{9} = 15.384888.$$

(3) For $d = 1 - c^2$,

$$\epsilon_2(c, 1-c^2) = 15 - 29c^2 + 13c^4 + c^6 \leq 15.$$

Therefore, we yield

$$|H_{2,3}(g)| \leq \frac{10\sqrt{3} + 675}{15552} = 0.044516.$$

The proof of Theorem 2 is completed. □

Theorem 3. *If $g \in \mathcal{B}T_{4l}$, then*

$$|H_{3,1}(g)| \leq \frac{3025}{46656} = 0.064836.$$

Proof. Assume that $g \in \mathcal{B}T_{4l}$. From (8) and (2), we achieve

$$\begin{aligned} |H_{3,1}(g)| &= \frac{1}{46656}\left|2250c_1c_2c_3 - 1000c_2^3 - 2025c_3^2 + 2160c_2c_4 - 1350c_1^2c_4\right| \\ &= \frac{1}{46656}\left|1000c_2(c_1c_3 - c_2^2) + 1250c_1c_2c_3 + 2025(c_2c_4 - c_3^2) + 135c_2c_4 - 1350c_1^2c_4\right| \end{aligned}$$

By applying the triangle inequality, Lemma 1 and Lemma 2 in (13), we receive

$$\begin{aligned} 46656|H_{3,1}(g)| &\leq 1000|c_2||c_1c_3 - c_2^2| + 1250|c_1||c_2||c_3| + 2025|c_2c_4 - c_3^2| + 135|c_2||c_4| + 1350|c_1|^2|c_4| \\ &\leq 1000|c_2|(1 - |c_1|^2) + 1250|c_1||c_2|(1 - |c_1|^2 - \frac{|c_2|^2}{1 + |c_1|}) + 2025(1 - |c_1|^2)^2 \\ &+135|c_2|(1 - |c_1|^2 - |c_2|^2) + 1350|c_1|^2(1 - |c_1|^2 - |c_2|^2) \\ &= 2025 - 2700|c_1|^2 + 675|c_1|^4 + (1135 + 1250|c_1| - 1135|c_1|^2 - 1250|c_1|^3)|c_2| \\ &-1350|c_1|^2|c_2|^2 - \frac{135 + 1385|c_1|}{1 + |c_1|}|c_2|^3. \end{aligned}$$

Setting $|c_1| = c$ and $|c_2| = d$, we have

$$\begin{aligned} 46656|H_{3,1}(g)| &\leq 2025 - 2700c^2 + 675c^4 + (1135 + 1250c - 1135c^2 - 1250c^3)d \\ &\quad -1350c^2d^2 - \frac{135 + 1385c}{1 + c}d^3 = \epsilon_3(c, d) \end{aligned}$$

where $c \in [0, 1]$ and $d \in [0, 1 - c^2]$.

Taking the partial derivative with respect to c and y respectively, and we have

$$\frac{\partial \epsilon_3}{\partial c} = -5400c + 2700c^3 + (1250 - 2270c - 3405c^2)d - 2700cd^2 - \frac{1250}{(1 + c)^2}d^3$$

and

$$\frac{\partial \epsilon_3}{\partial d} = 1135 + 1250c - 1135c^2 - 1250c^3 - 2700c^2d - \frac{405 + 4155c}{1 + c}d^2.$$

Setting $\frac{\partial \epsilon_3}{\partial c} = \frac{\partial \epsilon_3}{\partial d} = 0$ and simplifying, we yield

$$\begin{cases} -5400c - 10800c^2 - 2700c^3 + 5400c^4 + 2700c^5 + (1250 + 230c - 6695c^2 - 9080c^3 - 3405c^4)d \\ -2700c(1 + c)^2d^2 - 1250d^3 = 0, \\ 1135 + 2358c + 115c^2 - 2385c^3 - 1250c^4 - 2700c^2(1 + c)d - (405 + 4155c)d^2 = 0. \end{cases}$$

Applying Newton's methods, we yield

$$\begin{cases} c_1 = -1, \\ d_1 = 0, \end{cases} \quad \begin{cases} c_2 = 0.1018, \\ d_2 = -1.3080, \end{cases} \quad \begin{cases} c_3 = 1.0726, \\ d_3 = -1.1909, \end{cases} \quad \begin{cases} c_4 = 1.7878 \\ d_4 = -0.3706, \end{cases} \quad \begin{cases} c_5 = 3.3060, \\ d_5 = -6.5565, \end{cases} \quad \begin{cases} c_6 = -1.3977, \\ d_6 = 0.0899. \end{cases}$$

Thus, there is no critical point in $(0, 1) \times (0, 1 - c^2)$.

For $c = 0$,

$$\epsilon_3(0, d) = 2025 + 1135d - 135d^3 \leq \epsilon_3(0, 1) = 3025.$$

For $d = 0$,

$$\epsilon_3(c, 0) = 2025 - 2700c^2 + 675c^4 \leq 2025.$$

For $d = 1 - c^2$,

$$\epsilon_3(c, 0) = 3025 - 4665c^2 + 1605c^4 + 35c^6 \leq \epsilon_3(0, 0) = 3025.$$

Thus, we obtain

$$|H_{3,1}(g)| \leq \frac{3025}{46656} = 0.064836.$$

The proof of Theorem 3 is completed. □

Theorem 4. *If $g \in \mathcal{BT}_{4l}$, then*

$$|b_2b_5 - b_3b_4| \leq \frac{15.503407}{432} = 0.035888.$$

Proof. Let $g \in \mathcal{BT}_{4l}$. From (8), we obtain

$$|b_2b_5 - b_3b_4| = \frac{1}{432}|30c_1c_4 - 25c_2c_3|. \tag{13}$$

Applying Lemma 1 and the triangle inequality in (14), we get

$$|b_2b_5 - b_3b_4| \leq \frac{1}{432}[30|c_1|(1-|c_1|^2-|c_2|^2)+25|c_2|(1-|c_1|^2-\frac{|c_2|^2}{1+|c_1|})]$$
$$= \frac{1}{432}\left[30|c_1|-30|c_1|^3+(25-25|c_1|^2)|c_2|-30|c_1||c_2|^2-\frac{25}{1+|c_1|}|c_2|^3\right].$$

Setting $|c_1| = c$ and$|c_2| = d$, we yield

$$|b_2b_5 - b_3b_4| \leq \frac{1}{432}\epsilon_4(c,d).$$

Taking the partial derivative with respect to c and d, respectively, we obtain

$$\frac{\partial \epsilon_4}{\partial c} = 30 - 90c^2 - 50cd - 30d^2 + \frac{25}{(1+c)^2}d^3$$

and

$$\frac{\partial \epsilon_4}{\partial d} = 25 - 25c^2 - 60cd - \frac{75}{1+c}d^2.$$

Setting $\frac{\partial \epsilon_4}{\partial c} = \frac{\partial \epsilon_4}{\partial d} = 0$,and simplifying, we receive

$$\begin{cases} -90c^4 - 180c^3 - 60c^2 + 60c + 30 - 50c(1+c)^2d - 30(1+c)^2d^2 + 25d^3 = 0, \\ -25c^3 - 25c^2 + 25c + 25 - 60(c^2+c)d - 75d^2 = 0. \end{cases}$$

Applying Newton's methods, we have

$$\begin{cases} c_1 = -1, \\ d_1 = 0, \end{cases} \quad \begin{cases} c_2 = 0.4098, \\ d_2 = -0.8978, \end{cases} \quad \begin{cases} c_3 = 0.4271, \\ d_3 = 0.4258, \end{cases} \quad \begin{cases} c_4 = -0.4550 \\ d_4 = -0.2931, \end{cases} \quad \begin{cases} c_5 = -0.7258, \\ d_5 = 0.3023. \end{cases}$$

Therefore, we obtain $\varepsilon_4(c,d) \leq \varepsilon_4(0.4271, 0.4258) = 15.503407$.

(a) For $c = 0$,

$$\varepsilon_4(0,d) = 25d - 25d^3 \leq \varepsilon_4(0, \frac{\sqrt{3}}{3}) = \frac{50\sqrt{3}}{9}.$$

(b) For $d = 0$,

$$\varepsilon_4(c,0) = 30c - 30c^3 \leq \varepsilon_4(\frac{\sqrt{3}}{3}, 0) = \frac{20\sqrt{3}}{3} = 11.546666.$$

(c) For $d = 1 - c^2$,

$$\varepsilon_4(c, 1-c^2) = 25c - 20c^3 - 5c^5 = \eta_2(c) = \eta_2(0.6017) = 10.2913.$$

Thus, we obtain

$$|b_2b_5 - b_3b_4| \leq \frac{15.503407}{432} = 0.035888.$$

□

Theorem 5. *If $g \in \mathcal{B}T_{4l}$, then*

$$|H_{2,2}(g)| = |b_2b_4 - b_3^2| \leq \frac{200.27}{2592} = 0.077265.$$

Proof. Let $g \in \mathcal{B}T_{4l}$. From (8), we obtain

$$|b_2b_4 - b_3^2| = \frac{1}{2592}|225c_1c_3 - 200c_2^2|$$
$$= \frac{1}{2592}|200(c_1c_3 - c_2^2) + 25c_1c_3|.$$

Applying Lemma 1, Lemma 2 and the triangle inequality, we receive

$$\begin{aligned}|b_2b_4 - b_3^2| &\leq \frac{1}{2592}[200(1-|c_1|^2) + 25|c_1|(1-|c_1|^2 - \frac{|c_2|^2}{1+|c_1|})] \\ &= \frac{1}{2592}\left[200 + 25|c_1| - 200|c_1|^2 - 25|c_1|^3 - \frac{|c_1|}{1+|c_1|}|c_2|^2\right] \\ &\leq \frac{1}{2592}\left(200 + 25|c_1| - 200|c_1|^2 - 25|c_1|^3\right) \\ &= \frac{1}{2592}\eta_3(|c_1|) \leq \frac{1}{2592}\eta_3(0.0618) = \frac{200.2700}{2592} = 0.077265.\end{aligned}$$

□

Theorem 6. *If $g \in \mathcal{BT}_{4l}$, then*

$$|H_{3,2}(g)| \leq 0.025986.$$

Proof. Let $g \in \mathcal{BT}_{4l}$. From Lemma 3, Theorems 1, 2, 4 and 5, we yield

$$\begin{aligned}|H_{3,2}(g)| &\leq |b_4||b_3b_5 - b_4^2| + |b_5||b_2b_5 - b_3b_4| + |b_6||b_2b_4 - b_3^2| \\ &= \frac{5}{24} \times 0.044516 + \frac{1}{6} \times 0.035888 + \frac{5}{36} \times 0.077265 \\ &= 0.025986.\end{aligned}$$

□

Theorem 7. *If $g \in \mathcal{BT}_{4l}$, then*

$$|H_{2,4}(g)| = |b_4b_6 - b_5^2| \leq \frac{24.385252}{864} = 0.028224.$$

Proof. Let $f \in \mathcal{BT}_{4l}$. From (8) and (9), we have

$$|b_4b_6 - b_5^2| = \frac{1}{864}|25c_3c_5 - 24c_4^2 + 5c_3c_1^5| = \frac{1}{864}|24(c_3c_5 - c_4^2) + c_3c_5 + 5c_3c_1^5|.$$

Using Lemmas 1 and 2, we have

$$\begin{aligned}|b_4b_6 - b_5^2| &\leq \frac{1}{864}(24|c_3c_5 - c_4^2| + |c_3||c_5| + 5|c_3||c_1^5|) \\ &\leq \frac{1}{864}[24(1-|c_1|^2) + |c_3|(1-|c_1|^2 - |c_2|^2 - \frac{|c_3|^2}{1+||c_1}) + 5|c_1|^5|c_3|] \\ &= \frac{1}{864}[24 - 24|c_1|^2 + (1-|c_1|^2 + 5|c_1|^5 - |c_2|^2)|c_3| - \frac{1}{1+|c_1|}|c_3|^3].\end{aligned}$$

Setting $c = |c_1|, d = |c_2|$ and $e = |c_3|$, we yield

$$|b_4b_6 - b_5^2| \leq \frac{1}{864}\epsilon_5(c,d,e)$$

where $\epsilon_5(c,d,e) = 24 - 24c^2 + (1 - c^2 + 5c^5 - d^2)e - \frac{1}{1+c}e^3$ and $(c,d,e) \in \Omega = \{(c,d,e) : 0 \leq c \leq 1, 0 \leq d \leq 1-c^2, 0 \leq e \leq 1 - c^2 - \frac{d^2}{1+c}\}$. Consider

$$\frac{\partial \epsilon_5}{\partial d} = -2de \leq 0,$$

thus, there are no points in Ω.

(1) For $c = 0$,

$$\epsilon_5(0, d, e) = 24 + (1 - d^2)e - e^3 = \eta_4(d, e).$$

It is evident that there is on point in $(0, 1) \times (0, 1 - d^2)$.

(2) For $d = 0$,

$$\epsilon_5(c, 0, e) = 24 - 24c^2 + (1 - c^2 + 5c^5)e - \frac{1}{1+c}e^3 = \eta_5(c, e).$$

Partial derivative of η_5 with respect to c, and then with respect to e, we achieve

$$\frac{\partial \eta_5}{\partial c} = -48c + (-2c + 25c^4)e + \frac{1}{(1+c)^2}e^3,$$

and

$$\frac{\partial \eta_5}{\partial e} = 1 - c^2 + 5c^5 - \frac{3}{1+c}e^2.$$

Setting $\frac{\partial \eta_5}{\partial c} = \frac{\partial \eta_5}{\partial e} = 0$ and simplifying, we yield

$$\begin{cases} -48c(1+c)^2 + (25c^6 + 50c^5 + 25c^4 - 2c^3 - 4c^2 - 2c)e + e^3 = 0, \\ 5c^6 + 5c^5 - c^3 - c^2 + c + 1 - 3e^2 = 0. \end{cases}$$

We obtain a critical point $(0.0039, 0.5785)$; thus, we have $\eta_5(c, e) \leq \eta_5(0.0039, 0.5785) = 24.385252$.

(3) For $e = 0$,

$$\epsilon_5(c, d, 0) = 24 - 24c^2 \leq 24.$$

(4) For $e = 1 - c^2 - \frac{d^2}{1+c}$,

$$\begin{aligned} \epsilon_5(c, d, 1 - c^2 - \tfrac{d^2}{1+c}) &= 24 + c - 24c^2 - 2c^3 + 6c^5 - 5c^7 + \tfrac{-5c^5+4c^3-c^2-4c+1}{1+c}d^2 \\ &\quad + \tfrac{-2+4c}{(1+c)^2}d^4 + \tfrac{1}{(1+c)^4}d^6 = \eta_6(c, d) \leq \eta_6(0.0016, 0.5767) = 24.148169. \end{aligned}$$

(5) For $c = d = 0$

$$\epsilon_5(0, 0, e) = 24 + e - e^3 = \eta_7(e) \leq \eta_7(\frac{\sqrt{3}}{3}) = 24 + \frac{2\sqrt{3}}{9} = 24.384889.$$

(6) For $c = e = 0$,

$$\epsilon_5(0, d, 0) = 24.$$

(7) For $d = e = 0$,

$$\epsilon_5(c, 0, 0) = 24 + e - e^3 \leq 24.384889.$$

(8) For $e = 0, d = 1 - c^2$,

$$\epsilon_5(c, 1 - c^2, 0) = 24 - 24c^2 \leq 24.$$

(9) For $d = 0$ and $e = 1 - c^2$,

$$\epsilon_5(c, 0, 1 - c^2) = 24 + c - 24c^2 - 2c^3 + 6c^5 - 5c^7 = \eta_8(c) \leq \eta_8(0.0206) = 24.0104.$$

(10) For $c = 0, e = 1 - d^2$

$$\epsilon_5(0, d, 1 - d^2) = 24 + d^2 - 2d^4 + d^6 = \eta_9(d) \leq \eta_9(\frac{\sqrt{3}}{3}) = 24.148148.$$

Thus, we get

$$|b_4 b_6 - b_5^2| \leq \frac{24.385252}{864} = 0.035888.$$

□

Theorem 8. *If $g \in \mathcal{BT}_{4l}$, then*

$$|b_3 b_6 - b_4 b_5| \leq \frac{5}{144}.$$

Proof. Let $g \in \mathcal{BT}_{4l}$. From (8) and (9), we receive

$$|b_3 b_6 - b_4 b_5| = \frac{1}{1296}|50c_2c_5 - 45c_3c_4 + 50c_2c_1^5|.$$

Using Lemma 1, we obtain

$$\begin{aligned}
|b_3 b_6 - b_4 b_5| &\leq \frac{1}{1296}(50|c_2||c_5| + 45|c_3||c_4| + 50|c_2||c_1^5|) \\
&\leq \frac{1}{1296}[50|c_2|(1 - |c_1|^2 - |c_2|^2 - \frac{|c_3|^2}{1 + |c_1|}) + 45|c_3|(1 - |c_1|^2 - |c_2|^2) + 50|c_1|^5|c_2|] \\
&= \frac{1}{1296}[(50 - 50|c_1|^2 + 50|c_1|^5)|c_2| - 50|c_2|^3 + (45 - 45|c_1|^2 - 45|c_2|^2)|c_3| - \frac{50|c_2|}{1 + |c_1|}|c_3|^2].
\end{aligned}$$

By setting $|c_1| = c, |c_2| = d$ and $|c_3| = e$, we have

$$|b_3 b_6 - b_4 b_5| \leq \frac{1}{1296}\Psi(c, d, e)$$

where $\Psi(c, d, e) = (50 - 50c^2 + 50c^5)d - 50d^3 + (45 - 45c^2 - 45d^2)e - \frac{50d}{1+c}e^2$, $(c, d, e) \in \Omega$. Differentiating partially with respect to c, d and e, respectively, we obtain

$$\frac{\partial \Psi}{\partial c} = (-100c + 250c^4)d - 90ce + \frac{50d}{(1 + c)^2}e^2,$$

$$\frac{\partial \Psi}{\partial d} = 50 - 50c^2 + 50c^5 - 150d^2 - 90de - \frac{50}{1 + c}e^2,$$

and

$$\frac{\partial \Psi}{\partial e} = 45 - 45c^2 - 45d^2 - \frac{100d}{1 + c}e.$$

By putting $\frac{\partial \Psi}{\partial c} = \frac{\partial \Psi}{\partial d} = \frac{\partial \Psi}{\partial e} = 0$, and simplifying, we obtain

$$\begin{cases}
(250c^6 + 500c^5 + 250c^4 - 100c^3 - 200c^2 - 100c)d - 900c(1 + c)^2e + 50de^2 = 0, \\
50c^6 + 50c^5 - 50c^3 - 50c^2 + 50c + 50 - 150(1 + c)d^2 - 90(1 + c)de - 50e^2 = 0, \\
-45c^3 - 45c^2 + 45c + 45 - 45(1 + c)d^2 - 100de = 0.
\end{cases}$$

By a numberical calculation, we obtain

$$\begin{cases} c_1 = -1, \\ d_1 = d, \\ e_1 = 0, \end{cases} \quad \begin{cases} c_2 = 0.8441, \\ d_2 = 0.4127, \\ e_2 = 0.2354, \end{cases} \quad \begin{cases} c_3 = 0.8441, \\ d_3 = -0.4127, \\ e_3 = -0.2354. \end{cases}$$

Thus, there is no critical point that satisfies $0 \le c \le 1$ and $0 \le d \le 1 - c^2$.

(1) For $c = 0$,

$$\Psi(0, d, e) = 50d - 50d^3 + (45 - 45d^2)e - 50de^2 = \Lambda_1(d, e).$$

Consider

$$\begin{cases} \frac{\partial \Lambda_1}{\partial d} = 50 - 150d^2 - 90de - 50e^2 = 0, \\ \frac{\partial \Lambda_1}{\partial e} = 45 - 45d^2 - 100de = 0. \end{cases}$$

A numerical calculation is that there is no critical point in $(0, 1) \times (0, 1 - d^2)$.

(2) For $d = 0$,

$$\Psi(c, 0, e) = (45 - 45c^2)e \le 45(1 - c^2)^2 \le 45.$$

(3) For $e = 0$,

$$\Psi(c, d, 0) = (50 - 50c^2 + 50c^5)d - 50d^3 = \Lambda_2(c, d) \le \Lambda_2(0.7368, 0.2828) = 8.403278.$$

(4) For $e = 1 - c^2 - \frac{d^2}{1+c}$,

$$\begin{aligned} \Psi(c, d, 1 - c^2 - \frac{d^2}{1+c}) &= 45 - 90c^2 + 45c^4 + (50c - 50c^3 + 50c^5)d + (-90 + 45c + 45c^2)d^2 \\ &\quad + \frac{50 - 150c}{1+c}d^3 + \frac{45}{1+c}d^4 - \frac{50}{(1+c)^3}d^5 = \Lambda_3(c, d). \end{aligned}$$

Differentiating Λ_3 partially with respect to c and d, we yield

$$\begin{aligned} \frac{\partial \Lambda_3}{\partial c} &= -180c + 180c^3 + (50 - 150c^2 + 250c^4)d + (45 + 90c)d^2 - \frac{100d^3}{(1+c)^2} \\ &\quad - \frac{45d^4}{(1+c)^2} + \frac{150d^5}{(1+c)^4} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial \Lambda_3}{\partial d} &= 50c - 50c^3 + 50c^5 + (-180 + 90c + 90c^2)d + \frac{150 - 450c}{1+c}d^2 + \frac{180d^3}{1+c} \\ &\quad - \frac{250d^4}{(1+c)^3}. \end{aligned}$$

Setting $\frac{\partial \Lambda_3}{\partial c} = \frac{\partial \Lambda_3}{\partial d} = 0$ and simplifying, we receive

$$\begin{cases} 180c^7 + 720c^6 + 900c^5 - 900c^3 - 720c^2 - 180c + (250c^8 + 1000c^7 + 1350c^6 + 400c^5 \\ -600c^4 - 400c^3 + 150c^2 + 200c + 50)d + (90c^5 + 405c^4 + 720c^3 + 630c^2 + 270c + 45)d^2 \\ -100(1+c)^2d^3 - 45(1+c)^2d^4 + 150d^5 = 0, \\ 50c^8 + 150c^7 + 100c^6 - 100c^5 - 100c^4 + 100c^3 + 150c^2 + 50c + (90c^5 + 360c^4 + 360c^3 - 180c^2 \\ -450c - 180)d + (-450c^3 - 750c^2 - 150c + 150)d^2 + 180(1+c)^2d^3 - 250d^4 = 0 \end{cases}$$

Applying Newton's methods, we recieve

$$\begin{cases} c_1 = -1 \\ d_1 = 0, \end{cases} \quad \begin{cases} c_2 = 0, \\ d_2 = 0, \end{cases} \quad \begin{cases} c_3 = 0.8336, \\ d_3 = 0.4141, \end{cases} \quad \begin{cases} c_4 = -0.9194, \\ d_4 = -0.0957. \end{cases}$$

Thus, there is no critical point satisfing $0 \leq c \leq 1$ and $0 \leq d \leq 1 - c^2$.

(5) For $d = c = 0$,

$$\Psi(0, 0, e) = 45e \leq 45.$$

(6) For $e = d = 0$,

$$\Psi(c, 0, 0) = 0.$$

(7) For $c = e = 0$,

$$\Psi(0, d, 0) = 50d - 50d^3 = \Lambda_4(d) \leq \Lambda_4(\frac{\sqrt{3}}{3}) = \frac{100\sqrt{3}}{9} = 19.2444.$$

(8) For $d = 1 - c^2$ and $e = 0$,

$$\Psi(c, 1 - c^2, 0) = 50c^2 - 100c^4 + 50c^5 + 50c^6 - 50c^7 = \Lambda_5(c) \leq \Lambda_5(0.7145) = 10.6734.$$

(9) For $e = 1 - c^2$ and $d = 0$,

$$\Psi(c, 0, 1 - c^2) = 45(1 - c^2)^2 \leq 45.$$

(10) For $e = 1 - d^2$ and $c = 0$,

$$\Psi(0, d, 1 - c^2) = 45 - 90d^2 + 50d^3 + 45d^4 - 50d^5 = \Lambda_6(d) \leq \Lambda_6(0) = 45.$$

Hence, we obtain

$$|b_3 b_6 - b_4 b_5| \leq \frac{45}{1296} = \frac{5}{144}.$$

□

Theorem 9. *If $g \in \mathcal{BT}_{4l}$, then*

$$|H_{3,3}(g)| \leq 0.016103.$$

Proof. Let $g \in \mathcal{BT}_{4l}$. From Lemma 3 and Theorems 1–4, we receive

$$\begin{aligned} |H_{3,3}(g)| &\leq |b_5||b_4 b_6 - b_5^2| + |b_6||b_3 b_6 - b_4 b_5| + |b_7||b_3 b_5 - b_4^2| \\ &\leq \frac{1}{6} \times 0.035888 + \frac{5}{36} \times \frac{5}{144} + \frac{5}{42} \times 0.044516 \\ &= 0.016103. \end{aligned}$$

□

3. The Bounds of the Logarithmic Coefficients for $g \in \mathcal{BT}_{4l}$

Theorem 10. *If $g \in \mathcal{BT}_{4l}$, then*

$$|\delta_1| \leq \frac{5}{24}, \quad |\delta_2| \leq \frac{5}{36}, \quad |\delta_3| \leq \frac{5}{48}, \quad |\delta_4| \leq \frac{1}{12}, \quad |\delta_5| \leq \frac{173400}{2488320} = 0.069686.$$

The first four bounds are the best possible.

Proof. Let $g \in \mathcal{BT}_{4l}$. From(6), (8), and (9), we have

$$\delta_1 = \frac{5c_1}{24}, \tag{14}$$

$$\delta_2 = \frac{5}{36}(c_2 - \frac{5}{8}c_1^2), \tag{15}$$

$$\delta_3 = \frac{5}{48}(c_3 - \frac{5}{9}c_1c_2 + \frac{25}{216}c_1^3), \tag{16}$$

$$\delta_4 = \frac{1}{165888}(13824c_4 - 7200c_1c_3 + 4000c_1^2c_2 - 3200c_2^2 - 625c_1^4). \tag{17}$$

$$\delta_5 = \frac{1}{2488320}(172800c_5 - 86400c_1c_4 - 72000c_2c_3 + 45000c_1^2c_3 + 40000c_1c_2^2 - 5000c_1^3c_2 + 37685c_1^5). \tag{18}$$

Applying Lemma 4 to (15), we have

$$|\delta_1| \leq \frac{5}{24}$$

Applying Lemma 5 to (16), we obtain

$$|\delta_2| \leq \frac{5}{36}.$$

Utilizing the triangle inequality and Lemma 6 with $\gamma = -\frac{5}{9}$ and $\zeta = \frac{25}{216}$, we yield

$$|\delta_3| \leq \frac{5}{48}.$$

Rearranging (18), we obtain

$$\begin{aligned}
|\delta_4| &= \frac{1}{165888}|7200(c_4 - c_1c_3 - \frac{1}{2}c_2^2 + \frac{3}{4}c_1^2c_2 - \frac{1}{8}c_1^4) + 6624c_4 + 400c_2^2 - 1400c_1^2c_2 + 275c_1^4| \\
&\leq \frac{1}{165888}|7200(c_4 - c_1c_3 - \frac{1}{2}c_2^2 + \frac{3}{4}c_1^2c_2 - \frac{1}{8}c_1^4)| + \frac{1}{165888}|6624c_4 + 400c_2^2 - 1400c_1^2c_2 + 275c_1^4| \\
&= \frac{1}{165888}\nabla_1 + \frac{1}{165888}\nabla_2,
\end{aligned}$$

where $\nabla_1 = |7200(c_4 - c_1c_3 - \frac{1}{2}c_2^2 + \frac{3}{4}c_1^2c_2 - \frac{1}{8}c_1^4)|$

$$\nabla_2 = |6624c_4 + 400c_2^2 - 1400c_1^2c_2 + 275c_1^4|. \tag{19}$$

Using Lemma 7 with $\gamma = -\frac{1}{2}$, we obtain $\nabla_1 \leq 7200$.
Using Lemma 6 and the triangle inequality in (20), we receive

$$\begin{aligned}
\nabla_2 &\leq 6624(1 - |c_1|^2 - |c_2|^2) + 400|c_2|^2 + 1400|c_1|^2|c_2| + 275|c_1|^4 \\
&= 6624 - 6624|c_1|^2 + 275|c_1|^4 + 1400|c_1|^2|c_2| - 6224|c_2|^2 = \mathrm{Y}_1(c,d)
\end{aligned}$$

where $|c_1| = c, |c_2| = d$.
Consider

$$\begin{cases} \frac{\partial \mathrm{Y}_1}{\partial c} = -13248c + 1100c^3 + 2800cd = 0, \\ \frac{\partial \mathrm{Y}_1}{\partial d} = 1400c^2 - 12448d = 0 \end{cases}$$

Applying Newton's methods, we have

$$\begin{cases} c_1 = 0 \\ d_1 = 0, \end{cases} \quad \begin{cases} c_2 = 3.0599, \\ d_2 = 1.0531, \end{cases} \quad \begin{cases} c_3 = -3.0599, \\ d_3 = 1.0531. \end{cases}$$

Thus, in $(0,1) \times (0, 1-c^2)$, there is no critical point.

(1) For $c = 0$,

$$\mathrm{Y}_1(0,d) = 6624 - 6224d^2 \leq 6624.$$

(2) For $d = 0$,

$$\mathrm{Y}_1(c,0) = 6624 - 6624c^2 + 275c^4 = \gamma_1(c) \leq \gamma_1(0) = 6624.$$

(3) For $d = 1-c^2$,

$$\mathrm{Y}_1(c,1-c^2) = 400 + 7224c^2 - 7349c^4 = \gamma_2(c) \leq \gamma_2(0.7011) = 2175.3.$$

Therefore, we have

$$|\gamma_4| \leq \frac{13824}{165888} = \frac{1}{12}.$$

Rearranging (19), we obtain

$$\begin{aligned} |\delta_5| \quad &= \frac{1}{2488320}|86400(c_5 - c_1c_4 - c_2c_3 + \frac{3}{4}c_1c_2^2 + \frac{3}{4}c_1^2c_3 - \frac{1}{2}c_1^3c_2 + \frac{1}{16}c_1^5) \\ &+86400c_5 + 14400c_2c_3 - 19800c_1^2c_3 - 24800c_1c_2^2 + 38200c_1^3c_2 + 32285c_1^5| \\ &\leq \frac{1}{2488320}|86400(c_5 - c_1c_4 - c_2c_3 + \frac{3}{4}c_1c_2^2 + \frac{3}{4}c_1^2c_3 - \frac{1}{2}c_1^3c_2 + \frac{1}{16}c_1^5)| \\ &+\frac{1}{2488320}|86400c_5 + 14400c_2c_3 - 19800c_1^2c_3 - 24800c_1c_2^2 + 38200c_1^3c_2 + 32285c_1^5| \\ &= \frac{1}{2488320}\nabla_3 + \frac{1}{2488320}\nabla_4, \end{aligned}$$

where $\nabla_3 = |86400(c_5 - c_1c_4 - c_2c_3 + \frac{3}{4}c_1c_2^2 + \frac{3}{4}c_1^2c_3 - \frac{1}{2}c_1^3c_2 + \frac{1}{16}c_1^5)|$ and

$$\nabla_4 \quad = |86400c_5 + 14400c_2c_3 - 19800c_1^2c_3 - 24800c_1c_2^2 + 38200c_1^3c_2 + 32285c_1^5|. \tag{20}$$

Using Lemma 8 with $\gamma = -\frac{1}{2}$, we obtain $\nabla_3 \leq 86400$. Rearranging (21), we obtain

$$\nabla_4 \quad = |86400c_5 + 14400c_2(c_3 - 2c_1c_2 + c_1^3) - 19800c_1^2(c_3 - c_1c_2 - c_1^3) + 4000c_1c_2^2 + 4000c_1^3c_2 + 12485c_1^5|.$$

Utilizing the triangle inequality, Lemma 1, Lemma 6 and 5, we obtain

$$\begin{aligned} \nabla 4 \quad &\leq 86400|c_5| + 14400|c_2||c_3 - 2c_1c_2 + c_1^3| + 19800|c_1|^2|c_3 - c_1c_2 - c_1^3| + 4000|c_1||c_2|^2 + 4000|c_1|^3|c_2| + 12485|c_1|^5 \\ &\leq 86400(1 - |c_1|^2 - |c_2|^2 - \frac{|c_3|^2}{1+|c_1|}) + 14400|c_2| + 19800|c_1|^2 + 4000|c_1||c_2|^2 + 4000|c_1|^3|c_2| + 12485|c_1|^5 \\ &= 86400 - 666000|c_1|^2 + 12485|c_1|^5 + (14400 + 4000|c_1|^3)|c_2| + (-86400 + 4000|c_1|)|c_2|^2 - 86400\frac{|c_3|^2}{1+|c_1|} \\ &\leq 86400 - 666000|c_1|^2 + 12485|c_1|^5 + (14400 + 4000|c_1|^3)|c_2| + (-86400 + 4000|c_1|)|c_2|^2 = \mathrm{Y}_2(c,d) \end{aligned}$$

where $|c_1| = c$, $|c_2| = d$. Consider

$$\begin{cases} \frac{\partial Y_2}{\partial c} = -133200c + 62425c^4 + 12000c^2d + 4000d^2 = 0, \\ \frac{\partial Y_2}{\partial d} = 14400 + 4000c^3 + (-172800 + 8000c)d = 0. \end{cases}$$

We have a critical point $(0.000209, 0.083334)$. Thus, we obtain $Y_2(c,d) \leq Y_2(0.000209, 0.083334) = 86999.968$.

(1) For $c = 0$,

$$Y_2(0,d) = 86400 + 14400d - 86400d^2 = \gamma_3(d) \leq \gamma_3(\frac{1}{12}) = 87000.$$

(2) For $d = 0$,

$$Y_2(c,0) = 86400 - 66600c^2 + 12485c^5 \leq 864000.$$

(3) For $d = 1 - c^2$,

$$Y_2(c, 1-c^2) = 14400 + 4000c + 91800c^2 - 4000c^3 - 86400c^4 + 12485c^5 = \gamma_4(c) \leq \gamma_4(0.7771) = 43098.$$

Therefore, we receive

$$|\delta_5| \leq \frac{173400}{2488320} = 0.069686.$$

These bounds can be determined by the following extremal functions:

$$g_4(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t + \frac{1}{6}t^5)dt = \xi + \frac{5}{12}\xi^2 + \frac{1}{36}\xi^6 + \cdots,$$

$$g_5(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t^2 + \frac{1}{6}t^{10})dt = \xi + \frac{5}{18}\xi^3 + \frac{1}{66}\xi^{11} + \cdots,$$

$$g_6(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t^3 + \frac{1}{6}t^{15})dt = \xi + \frac{5}{24}\xi^4 + \frac{1}{96}\xi^{16} + \cdots,$$

$$g_7(\xi) = \int_0^{\xi} (1 + \frac{5}{6}t^4 + \frac{1}{6}t^{20})dt = \xi + \frac{1}{6}\xi^5 + \frac{1}{126}\xi^{21} + \cdots.$$

The proof of Theorem 10 is completed. □

4. Conclusions

In this paper, we considered a subclass of bounded turning functions linked with a four-leaf-type domain. Utilizing the estimates of the coefficients of the Schwartz function, we obtained the estimates of coefficients of $|b_6|, |b_7|, |b_8|$, and the third-order determinants of $|H_{3,1}|, |H_{3,2}|, |H_{3,3}|$ for the class $\mathcal{B}T_{4l}$ for the first time. Also, one can easily use this new methodology to obtain the bounds the coefficients of $|b_6|, |b_7|, |b_8|$, and the third-order Hankel determinant of $|H_{3,1}|, |H_{3,2}|, |H_{3,3}|$ for other subclasses of univalent functions. The bounds provided in Theorems 2-9 lack sharpness and improve these bounds in further research.

Author Contributions: C.W.; Writing—original draft, Z.L.; Methodology; D.G.; Validation. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created in this study.

Conflicts of Interest: The authors state that they have no conflicts of interest.

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